

Hospital Management System

Short Testing Report - Task 2.9

1. Testing Objective

The purpose of this testing was to verify that the Hospital Management System creates the required sample data, demonstrates the required object-oriented programming concepts, rejects invalid input, and produces correct reports through the menu.

2. Test Results

Test	Observed Result	Status
Sample data	6 patients, 2 inpatients, 4 doctors, 3 nurses, 6 appointments, 5 medical records	PASS
Method overloads / utilities	Overloaded methods and HelperUtils tests executed successfully	PASS
Polymorphism	Different Person subclasses displayed correctly at runtime	PASS
Validation	Invalid ID, age, balance, shift and appointment status were rejected	PASS
Reports	Menu report showed the seeded totals and calculated values	PASS

3. Test Evidence

```
--- SEEDING SAMPLE DATA ---
Sample data seeded successfully.

=====
TASK 2.9 TESTS
=====
Patients: 6
InPatients: 2
Doctors: 4
Nurses: 3
Appointments: 6
Medical Records: 5
Reason: Annual fee review
generateId(): 1
generateId(prefix): TEST-2
isEmpty text: true
isEmpty array: true
validText: true
validText min: true
validText min/max: true
positive int: true
positive double: true
range int: true
range double: true
```

Figure 1. Sample data and utility/overload test output.

```
--- POLYMORPHISM TEST ---
ID: BASE01, Name: Test Person, Age: 36, Phone: 90000000, Email: base@email.com
Patient: Ahmed Ali, ID: P001, Blood Group: Unknown, Emergency Contact: N/A, Registration Date: N/A, Outstanding Balance: 0.0, Insured: false
Doctor: Omar Salim, ID: D001, Specialization: Cardiology, Experience: 15 years, Consultation Fee: 40.0, Patient Load: 1, On Call: true
Nurse: Maryam Said, ID: N001, Department ID: WARD01, Shift: Morning, Years of Service: 8, Patient Load: 1
InPatient: Fatma Ahmed, ID: P005, Blood Group: AB+, Admission Date: 19-08-2026, Room Number: R101, Daily Charges: 50.0, Days Admitted: 3, Total Room Cost: 150.0
Surgeon: Hassan Ahmed, ID: D004, Specialization: Surgery, Surgeries Performed: 150, Theatre Access: true, Upcoming Surgeries: 1
```

Figure 2. Polymorphism test using Person, Patient, Doctor, Nurse, InPatient and Surgeon.

```
--- VALIDATION TEST ---
ID cannot be empty.
Age must be between 0 and 120.
Outstanding balance cannot be negative.
Shift must be Morning, Evening, or Night.
Invalid appointment status.
```

Figure 3. Validation test showing invalid inputs rejected.

```
=====
HOSPITAL MANAGEMENT SYSTEM
=====
1. Patients
2. Doctors
3. Nurses
4. Appointments
5. Medical Records
6. Reports
7. Exit
Choose option: 6

=====
HOSPITAL REPORTS
=====
Total Patients: 6
Total InPatients: 2
Total Doctors: 4
Total Nurses: 3
Total Appointments: 6
Total Medical Records: 5
Confidential Records: 3
Total Outstanding Balance: 1070.0
=====
```

Figure 4. Hospital Reports generated from the main menu.

4. Conclusion

The required Task 2.9 sample data and tests completed successfully. The console evidence shows correct sample-data counts, successful polymorphic behavior, working validation, and correct hospital report totals. Overall testing result: **PASS**.